

Math-2108 (LIDE)

Elementary Number Theory

The Division Algorithm

Let a be any integer and b a positive integer. Then there exist unique integers q and r such that $a = bq + r$ where $0 \leq r < b$.

$$\begin{array}{r} a \\ b \overline{) a} \\ \hline \end{array}$$

$$a = bq + r$$

a = dividend, b = divisor,
 r = remainder, q = quotient

Ex 1 Let r be the remainder when 1059, 1417 and 2312 are divided by $d > 1$. Find the value of $d - r$.

Solution

By the Division Algorithm,

$$1059 = q_1 d + r$$

$$1417 = q_2 d + r$$

$$2312 = q_3 d + r, \text{ for some integers } q_1, q_2, q_3.$$

$$358 = 1417 - 1059 = d(q_2 - q_1)$$

$$1253 = 2312 - 1059 = d(q_3 - q_1) \text{ and}$$

$$895 = 2312 - 1417 = d(q_3 - q_2).$$

$$d | 358 = 2 \cdot 179$$

$$d | 1253 = 7 \cdot 179 \text{ and}$$

$$d | 895 = 5 \cdot 179.$$

Since $d > 1$, $d = 179$

$$\therefore 1059 = 5 \cdot 179 + 164$$

$$\therefore r = 164$$

$$\text{Thus } d - r = 179 - 164 = 15$$

Definition (1)

An integer a is said to be divisible by an integer $b \neq 0$, is symbolized $b | a$, if there exists some integer c such that $a = bc$. We write $b \nmid a$ to indicate that a is not divisible by b .

Definition (2)

Let a and b be given integers, with at least one of them different from zero.

The greatest common divisor of a and b , denoted by $\gcd(a, b)$, is the positive integer d satisfying the following:

(a) $d | a$ and $d | b$; and

(b) If $c | a$ and $c | b$, then $c \leq d$, where c is also a positive integer.

Definition (3)

A linear combination of the integers a and b is a sum of multipliers of a and b , that is, a sum of the form $\alpha a + \beta b$, where α and β are integers.

Definition (4)

The least common multiple of two nonzero integers a and b , denoted by $\text{lcm}(a, b)$, is the positive integer m satisfying the following:

(a) $a|m$ and $b|m$

(b) If $a|c$ and $b|c$, with $c > 0$, then $m \leq c$.

Theorem

For positive integers a and b , $\text{gcd}(a, b) \text{lcm}(a, b) = ab$.

No (2) Find the gcd and lcm of 306 and 657. Also express the gcd as a linear combination $ax + by$ in each case.

Solution

$$\text{gcd}(306, 657) = ?$$

$$657 = 306 \times 2 + 45$$

$$306 = 45 \times 6 + 36$$

$$45 = 36 \times 1 + 9$$

$$36 = 9 \times 4 + 0$$

$$\therefore \text{gcd}(306, 657) = 9$$

By theorem,

$$\text{gcd}(306, 657) \text{lcm}(306, 657) = 306 \times 657$$

$$\text{lcm}(306, 657) = \frac{306 \times 657}{\text{gcd}(306, 657)}$$

*gcd * lcm = a * b
lcm = a * b / gcd*

$$= \frac{306 \times 657}{9}$$

$$= 22338$$

$$\text{gcd}(306, 657) = 9$$

$$= 45 - 36$$

$$= 45 - [306 - 45 \times 6]$$

$$= 7 \times 45 - 306$$

$$= 7[657 - 306 \times 2] - 306$$

$$= 7(657) + (-15)306$$

$$= 306(-15) + 657(7) = ax + by,$$

where $a = 306, b = 657, x = -15, y = 7$.

Theorem

The linear Diophantine Equation $ax + by = c$ is solvable if and only if $d|c$, where $d = \text{gcd}(a, b)$. If x_0, y_0 is a particular solution of the linear DE, then all its solutions are given by $x = x_0 + \left(\frac{b}{d}\right)t$ and $y = y_0 - \left(\frac{a}{d}\right)t$ where t is an arbitrary integer.

2) Determine all solutions in the positive integers of the Diophantine equation $158x - 57y = 7$.

Solution

$$158x - 57y = 7$$

$$158 = 57 \times 2 + 44$$

$$57 = 44 \times 1 + 13$$

$$44 = 13 \times 3 + 5$$

$$13 = 5 \times 2 + 3$$

$$5 = 3 \times 1 + 2$$

$$3 = 2 \times 1 + 1$$

$$2 = 1 \times 2 + 0$$

$$\therefore \gcd(158, 57) = 1$$

Since $1 \mid 7$, the eqⁿ has a solution.

$$1 = 3 - 2$$

$$= 3 - [5 - 3]$$

$$= 2 \times 3 - 5$$

$$= 2 [13 - 5 \times 2] - 5$$

$$= 2 \times 13 + (-5)(5)$$

$$= 2 \times 13 + (-5) [44 - 13 \times 3]$$

$$= 17 \times 13 + (-5)(44)$$

$$= 17 [57 - 44] + (-5)(44)$$

$$= 17 \times 57 + (-22)(44)$$

$$= 17(57) + (-22)[158 - 57 \times 2]$$

$$= 61(57) + (-22)158$$

$$7 = 158(-154) - 57(-427)$$

$$\therefore x_0 = -154, y_0 = -427$$

$$x = x_0 + \frac{b}{d}t, y = y_0 - \frac{a}{d}t$$

$$x = -154 - 57t, y = -427 - 158t$$

For solutions in the positive integers

$$-154 - 57t > 0 \text{ and } -427 - 158t > 0$$

$$t < -\frac{154}{57} (\approx -2.70) \text{ and } t < -2.70$$

Therefore, all solutions in the positive integers will be $x = -154 - 57t$ and $y = -427 - 158t$ where $t \leq -3$.

The Euclidean Algorithm

$$a = q_1 b + r_1, 0 < r_1 < b$$

$$b = q_2 r_1 + r_2, 0 < r_2 < r_1$$

$$r_1 = q_3 r_2 + r_3, 0 < r_3 < r_2$$

$$\vdots$$

$$r_{n-2} = q_n r_{n-1} + r_n, 0 < r_n < r_{n-1}$$

$$r_{n-1} = q_{n+1} r_n + 0$$

$$\gcd(a, b) = r_n$$

Unit 2

Primes and their Distributions

Definition (1)

An integer $p > 1$ is called a prime number or simply a prime, if its only positive divisors are 1 and p . An integer greater than 1 that is not a prime is termed composite.

Theorem 2.1

If p is a prime and $p \mid ab$ then $p \mid a$ or $p \mid b$.

Corollary 2.2

If p is a prime and $p \mid a_1 a_2 \dots a_n$ then $p \mid a_k$ for some k , where $1 \leq k \leq n$.

Corollary 2.3

If $p, q_1, q_2, \dots, q_n$ are all primes and $p \mid q_1 q_2 \dots q_n$, then $p = q_k$ for some k , where $1 \leq k \leq n$.

Theorem 2.4 Every positive integer $n > 1$ is either a prime or a product of primes; this representation is unique, apart from the order in which the factors occur.

Corollary 2.5

Any positive integer $n > 1$ can be written uniquely in a canonical form

$$n = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$$

where, for $i = 1, 2, \dots, r$, each k_i is a positive integer and each p_i is a prime with

$$p_1 < p_2 < \dots < p_r.$$

~~No (4)~~

Prove that each integer of the form $3n+2$ has a prime factor of this form.

Proof:

Let N be an integer of the form $3n+2$.

$$\therefore N = 3n+2.$$

Assume that no prime factor of N is of the form $3k+2$.

(1) Since $N = 3n+2$, $3 \nmid N$.

Thus 3 is not a factor of N .

(2) Any prime number can be written in one of the form:

$$3k, 3k+1 \text{ or } 3k+2.$$

Thus, all prime factor of N must be of the form $3k+1$.

$N = p_1 p_2 \dots p_r$ where each

$$p_i = 3k_i + 1$$

Consider

$$(3k_1 + 1)(3k_2 + 1) = 9k_1 k_2 + 3k_1 + 3k_2 + 1$$

$$= 3(3k_1 k_2 + k_1 + k_2) + 1$$

$$= 3m + 1 \text{ where } m = 3k_1 k_2 + k_1 + k_2$$

By mathematical induction, the product of any number of primes of the form $3k+1$ is also of the form $3m+1$.

$$\therefore N = 3m + 1$$

$\therefore$ A number cannot have a remainder of both 1 and 2 when divided by 3.

So our assumption was wrong.

$\therefore N = 3n + 2$ must have at least one prime factor of the form $3k+2$.

(ii) Prove that the only prime of the form $n^2 - 4$ is 5.

Proof:

Let P be a prime number of the form $n^2 - 4$, where n is an integer.

$$P = n^2 - 4 = (n-2)(n+2)$$

P can only be factored as $1 \cdot P$ (or) $(-1)(-P)$.

Since $n-2 < n+2$, the smaller factor must equal 1 or larger factor

must equal -1.

$$\text{Case 1, } n-2 = 1$$

$$n = 3$$

$$P = 3^2 - 4 = 5$$

$$\text{Case 2, } n+2 = -1$$

$$n = -3$$

$$P = (-3)^2 - 4 = 5$$

In both valid case, the only prime number is 5.

Unit 3

The Theory of Congruences

Definition (1)

Let n be a fixed positive integer. Two integers a and b are said to be congruent modulo n , symbolized by

$$a \equiv b \pmod{n}$$

if n divides the difference $a-b$; that is, provided that $a-b = kn$ for some integer k .

Consider $n = 7$.

$$3 \equiv 24 \pmod{7}, -31 \equiv 11 \pmod{7}$$

$$\text{and } -15 \equiv -64 \pmod{7}$$

$$\text{because } 3 - 24 = -21 = (-3)7,$$

$$-31 - 11 = -42 = (-6)7 \text{ and}$$

$$-15 + 64 = 49 = 7(7).$$

No ⑤ Prove that the integer

$53^{103} + 103^{53}$ is divisible by 39, and that $111^{333} + 333^{111}$ is divisible by 7.

Proof:

$$39 = 3 \times 13 \text{ and } \gcd(3, 13) = 1$$

$$53 \equiv -1 \pmod{3}$$

$$53^{103} \equiv (-1)^{103} \equiv -1 \pmod{3}$$

$$103 \equiv 1 \pmod{3}$$

$$103^{53} \equiv 1^{53} \equiv 1 \pmod{3}$$

$$\therefore 53^{103} + 103^{53} \equiv -1 + 1 \equiv 0 \pmod{3}$$

$$\therefore 53^{103} + 103^{53} \text{ is divisible by 3.}$$

$$53 \equiv 1 \pmod{13}$$

$$53^{103} \equiv 1^{103} \equiv 1 \pmod{13}$$

$$103 \equiv -1 \pmod{13}$$

$$103^{53} \equiv (-1)^{53} \equiv -1 \pmod{13}$$

$$53^{103} + 103^{53} \equiv 1 + (-1) \equiv 0 \pmod{13}$$

$$\therefore 53^{103} + 103^{53} \text{ is divisible by 13.}$$

$$\therefore 53^{103} + 103^{53} \text{ is divisible by 39.}$$

$$111 \equiv -1 \pmod{7}$$

$$111^{333} \equiv (-1)^{333} \equiv -1 \pmod{7}$$

$$333 \equiv 4 \pmod{7}$$

By Fermat's Little Theorem, $4^6 \equiv 1 \pmod{7}$

$$4^3 \equiv 1 \pmod{7}$$

$$333^{111} \equiv 4^{111} \equiv (4^3)^{37} \cdot 4^3 \pmod{7}$$

$$333^{111} \equiv (1)^{37} \cdot 64 \pmod{7}$$

$$333^{111} \equiv 1 \pmod{7}$$

$$\therefore 111^{333} + 333^{111} \equiv -1 + 1 \equiv 0 \pmod{7}$$

$$\therefore 111^{333} + 333^{111} \text{ is divisible by 7.}$$

No ⑥ Find the solutions of the system of congruences:

$$3x + 4y \equiv 5 \pmod{13}$$

$$2x + 5y \equiv 7 \pmod{13}$$

Solution

$$3x + 4y \equiv 5 \pmod{13}$$

$$2x + 5y \equiv 7 \pmod{13}$$

Since $\gcd(15-8, 13) = \gcd(7, 13) = 1$, a solution exists.

$$6x + 8y \equiv 10 \pmod{13}$$

$$-6x + 2y \equiv -8 \pmod{13}$$

$$6y \equiv 2 \pmod{13}$$

$$66y \equiv 22 \pmod{13}$$

$$y \equiv 9 \pmod{13}$$

$$15x + 20y \equiv 25 \pmod{13}$$

$$8x + 20y \equiv 28 \pmod{13}$$

$$7x \equiv -3 \pmod{13}$$

$$7x \equiv 10 \pmod{13}$$

$$14x \equiv 20 \pmod{13}$$

$$x \equiv 7 \pmod{13}$$

The unique sol. of our system turns out to be $x \equiv 7 \pmod{13}$, $y \equiv 9 \pmod{13}$

① Find the remainder when $(n^2+n+41)^2$ is divided by 12.
 Solution

First, notice that the product of four consecutive integers is divisible by 12.

i.e;

$$(n-1)n(n+1)(n+2) \equiv 0 \pmod{12}.$$

$$(n^2+n+41)^2 \equiv (n^2+n+5)^2 \pmod{12}$$

$$\equiv (n^4 + 2n^3 + 10n^2 + 10n + 25) \pmod{12}$$

$$\equiv (n^4 + 2n^3 - n^2 - 2n) + 1 \pmod{12}$$

$$\equiv n(n^3 + 2n^2 - n - 2) + 1 \pmod{12}$$

$$\equiv n[n^2(n+2) - (n+2)] + 1 \pmod{12}$$

$$\equiv n(n+2)(n^2-1) + 1 \pmod{12}$$

$$\equiv (n-1)n(n+1)(n+2) + 1 \pmod{12}$$

$$\equiv 1 \pmod{12}$$

Thus, when $(n^2+n+41)^2$ is divided by 12, the remainder is 1.

Definition

A composite integer n is called pseudoprime whenever $n \mid 2^n - 2$.

* If $d \mid n$ then $2^d - 1 \mid 2^n - 1$

$$* x^k - 1 = (x-1)(x^{k-1} + x^{k-2} + \dots + x + 1)$$

No ⑧ If n is an odd pseudoprime prove that $M_n = 2^n - 1$ is a larger one.

Solution

Since n is a composite number, $n = rs$ with $1 < r \leq s < n$.

Since $r \mid n$, $2^r - 1 \mid 2^n - 1$.

$$\therefore 2^r - 1 \mid M_n$$

$\therefore M_n$ is composite.

Since n is an odd pseudoprime,

$$n \mid 2^n - 2$$

$$\therefore 2^n \equiv 2 \pmod{n}$$

$$2^n - 2 = Kn \text{ for some integer } K.$$

$$\frac{M_n - 1}{2} = \frac{2^n - 2}{2} = \frac{Kn}{2}$$

$$\frac{M_n - 1}{2} - 1 = \frac{Kn}{2} - 1$$

$$= (2^n - 1)(2^{n(K-1)} + 2^{n(K-2)} + \dots + 2^{n-1} + 1)$$

$$= M_n (2^{n(K-1)} + 2^{n(K-2)} + \dots + 2^{n-1} + 1)$$

$$\equiv 0 \pmod{M_n}$$

$$\therefore 2^{M_n} - 2 \equiv 0 \pmod{M_n}$$

$$\therefore M_n \mid 2^{M_n} - 2$$

$\therefore M_n$ is pseudoprime.

No(9) Use Fermat's theorem to verify that 17 divides $11^{104} + 1$.
solution

$$a=11, p=17$$

By Fermat's theorem, $a^{p-1} \equiv 1 \pmod{p}$

17 is prime, $17 \nmid 11$ then

$$11^{17-1} \equiv 1 \pmod{17}$$

$$11^{16} \equiv 1 \pmod{17}$$

$$(11^{16})^6 \equiv 1^6 \pmod{17}$$

$$11^{96} \equiv 1 \pmod{17}$$

$$11^{104} = 11^{96+8} = 11^{96} \cdot 11^8$$

$$11^1 = 11 \equiv -6 \pmod{17}$$

$$11^2 \equiv 36 \equiv 2 \pmod{17}$$

$$11^4 \equiv 4 \pmod{17}$$

$$11^8 \equiv 16 \equiv -1 \pmod{17}$$

$$11^{104} = 11^{96} \cdot 11^8 \equiv 1(-1) \equiv -1 \pmod{17}$$

$$11^{104} + 1 \equiv 0 \pmod{17}$$

$$\therefore 17 \mid 11^{104} + 1$$

Thus, 17 divides $11^{104} + 1$.

Definition

Given a positive integer n , let $\tau(n)$ denote the number of positive divisors of n and $\sigma(n)$ denote the sum of these divisors.

No(10) Show that $\sigma(n) = \sigma(n+1)$

when $n = 14, 206$ and 957 .

solution

$$\text{For } n=14, n+1=15$$

$$\sigma(n) = \sigma(14) = 1+2+7+14 = 24$$

$$\sigma(n+1) = \sigma(15) = 1+3+5+15 = 24$$

$\therefore$ When $n=14$, $\sigma(n) = \sigma(n+1)$.

$$\text{For } n=206, n+1=207$$

$$\sigma(n) = \sigma(206) = \sigma(2 \times 103)$$

$$= (1+2)(1+103) = 3(104) = 312$$

$$\sigma(n+1) = \sigma(207) = \sigma(3^2 \times 23)$$

$$= (1+3+9)(1+23) = 13(24) = 312$$

$\therefore$ When $n=206$, $\sigma(n) = \sigma(n+1)$

$$\text{For } n=957, n+1=958$$

$$\sigma(n) = \sigma(957) = \sigma(3 \times 11 \times 29)$$

$$= (1+3)(1+11)(1+29) = 4 \cdot 12 \cdot 30 = 1440$$

$$\sigma(n+1) = \sigma(958) = \sigma(2 \times 479)$$

$$= (1+2)(1+479) = 3(480) = 1440$$

When $n=957$, $\sigma(n) = \sigma(n+1)$.

The number of leap year

$$L = \left[\frac{Y}{4} \right] - \left[\frac{Y}{100} \right] + \left[\frac{Y}{400} \right] - 388$$

* The number of day

$$\omega \equiv d + [(2.6)m - 0.2] - 2c + y + \left[\frac{c}{4} \right] + \left[\frac{y}{4} \right] \pmod{7}$$

No ⑪ Find the number n of leap years such that $1600 < n < Y$, when

(a) $Y = 1825$ (b) $Y = 1950$.

Solution

(a) $Y = 1825$

$$L = \left[\frac{Y}{4} \right] - \left[\frac{Y}{100} \right] + \left[\frac{Y}{400} \right] - 388$$

$$= \left[\frac{1825}{4} \right] - \left[\frac{1825}{100} \right] + \left[\frac{1825}{400} \right] - 388$$

$$= 456 - 18 + 4 - 388$$

$$= 54$$

$\therefore$ The number of leap years such that $1600 < n < Y$ when

$Y = 1825$ is 54.

(b) $Y = 1950$

$$L = \left[\frac{Y}{4} \right] - \left[\frac{Y}{100} \right] + \left[\frac{Y}{400} \right] - 388$$

$$= \left[\frac{1950}{4} \right] - \left[\frac{1950}{100} \right] + \left[\frac{1950}{400} \right] - 388$$

$$= 487 - 19 + 4 - 388 = 84$$

The number of leap years such that $1600 < n < Y$, when $Y = 1950$ is 84.

No ⑫ Find the day of the week for (a) April 18, 1906 (San Francisco earthquake).
(b) November 11, 1918 (Great War ends)

Solution

(a) April 18, 1906 (San Francisco earthquake)

$d = 18, m = 2, c = 19, y = 6$

$$\omega \equiv d + [(2.6)m - 0.2] - 2c + y + \left[\frac{c}{4} \right]$$

$$+ \left[\frac{y}{4} \right] \pmod{7}$$

$$\equiv 18 + [(2.6)2 - 0.2] - 2(19) + 6 + \left[\frac{19}{4} \right]$$

$$+ \left[\frac{6}{4} \right] \pmod{7}$$

$$\equiv 18 + 5 - 38 + 6 + 4 + 1 \pmod{7}$$

$$\equiv -4 \pmod{7}$$

$$\equiv 3 \pmod{7}$$

$\therefore$ April 18, 1906 (San Francisco earthquake) is Wednesday.

(b) November 11, 1918 (Great War ends)

$d = 11, m = 9, c = 19, y = 18$

$$\omega \equiv d + [(2.6)m - 0.2] - 2c + y + \left[\frac{c}{4} \right] + \left[\frac{y}{4} \right] \pmod{7}$$

$$\equiv 11 + [(2.6)9 - 0.2] - 2(19) + 18 +$$

$$\left[\frac{19}{4} \right] + \left[\frac{18}{4} \right] \pmod{7}$$

$$\omega \equiv 11 + 23 - 38 + 18 + 4 + 4 \pmod{7}$$

$$\equiv 22 \pmod{7}$$

$$\equiv 1 \pmod{7}$$

$\therefore$ November 11, 1918 (Great War ends) is Monday.

* If the integer $n > 1$ has the prime factorization

$$n = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}, \text{ then}$$

$$\phi(n) = (p_1^{k_1} - p_1^{k_1-1}) (p_2^{k_2} - p_2^{k_2-1}) \dots (p_r^{k_r} - p_r^{k_r-1})$$

$$= n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_r}\right)$$

No (13) Calculate $\phi(1001)$, $\phi(5040)$, and $\phi(36,000)$.

Solution

$$1001 = 7 \times 11 \times 13$$

$$\phi(1001) = 1001 \left(1 - \frac{1}{7}\right) \left(1 - \frac{1}{11}\right) \left(1 - \frac{1}{13}\right)$$

$$= 1001 \left(\frac{6}{7}\right) \left(\frac{10}{11}\right) \left(\frac{12}{13}\right)$$

$$= 720$$

$$5040 = 2^4 \times 3^2 \times 5 \times 7$$

$$\phi(5040) = 5040 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{7}\right)$$

$$= 5040 \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) \left(\frac{4}{5}\right) \left(\frac{6}{7}\right)$$

$$= 1152$$

$$36000 = 36 \times 1000 = 2^3 \times 3^2 \times 5^3$$

$$\phi(36000) = 36000 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right)$$

$$= 36000 \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) \left(\frac{4}{5}\right)$$

$$= 9600$$

Definition

For $n \geq 1$, let $\phi(n)$ denote the number of positive integers not exceeding n that are relatively prime to n .

$$\text{Eg. } \phi(30) = 8,$$

$$1, 7, 11, 13, 17, 19, 23, 29.$$

No (14) Show that the integers $m = 3^k \cdot 568$ and $n = 3^k \cdot 638$, where $k \geq 0$, satisfy simultaneously

$$\tau(m) = \tau(n) \text{ and } \phi(m) = \phi(n).$$

Solution

$$m = 3^k \cdot 568 = 3^k \cdot 2^3 \cdot 71$$

$$n = 3^k \cdot 638 = 3^k \cdot 2 \cdot 11 \cdot 29$$

$$\tau(m) = \tau(3^k \cdot 2^3 \cdot 71)$$

$$= (k+1)(3+1)(1+1)$$

$$= 8(k+1)$$

$$\tau(n) = \tau(3^k \cdot 2 \cdot 11 \cdot 29)$$

$$= (k+1)(1+1)(1+1)(1+1)$$

$$= 8(k+1)$$

$$\therefore \varphi(m) = \varphi(n)$$

$$\varphi(m) = \varphi(3^K \cdot 2^3 \cdot 71)$$

$$= \varphi(3^K) \varphi(2^3) \varphi(71)$$

$$= 3^K (1 - \frac{1}{3}) 2^3 (1 - \frac{1}{2}) 71 (1 - \frac{1}{71})$$

$$= 3^K (\frac{2}{3}) \cdot 2^3 (\frac{1}{2}) \cdot 71 (\frac{70}{71})$$

$$= 3^{K-1} \cdot 560$$

$$\varphi(n) = \varphi(3^K \cdot 2 \cdot 11 \cdot 29)$$

$$= \varphi(3^K) \varphi(2) \varphi(11) \varphi(29)$$

$$= 3^K (1 - \frac{1}{3}) \cdot 2 (1 - \frac{1}{2}) \cdot 11 (1 - \frac{1}{11}) 29 (1 - \frac{1}{29})$$

$$= 3^K (\frac{2}{3}) \cdot 2 (\frac{1}{2}) \cdot 11 (\frac{10}{11}) \cdot 29 (\frac{28}{29})$$

$$= 3^{K-1} \cdot 560$$

$$\therefore \varphi(m) = \varphi(n)$$

* If $n = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$ is the prime factorization of $n > 1$, then

$$\varphi(n) = (k_1 + 1)(k_2 + 1) \dots (k_r + 1).$$

No (15) (i) Evaluate $\varphi(221)$ and $\varphi(6125)$

(ii) Find the remainder when

245^{1040} is divided by 18.

Solution

$$(i) \varphi(221) = \varphi(13 \cdot 17)$$

$$= \varphi(13) \varphi(17)$$

$$= 13 (1 - \frac{1}{13}) 17 (1 - \frac{1}{17})$$

$$= 13 (\frac{12}{13}) 17 (\frac{16}{17})$$

$$= 12 \cdot 16$$

$$= 192$$

$$\varphi(6125) = \varphi(5^3 \cdot 7^2)$$

$$= \varphi(5^3) \varphi(7^2)$$

$$= 5^3 (1 - \frac{1}{5}) 7^2 (1 - \frac{1}{7})$$

$$= 5^3 (\frac{4}{5}) \cdot 7^2 (\frac{6}{7})$$

$$= 100 \cdot 42$$

$$= 4200$$

(ii) Since $245 \equiv 11 \pmod{18}$,

$$245^{1040} \equiv 11^{1040} \pmod{18}$$

Since $\gcd(11, 18) = 1$,

$$11^{\varphi(18)} \equiv 1 \pmod{18}$$

Therefore, $11^{1040} \equiv (11^6)^{173} \cdot 11^2$

$$\equiv (1)^{173} \cdot 13 \equiv 13 \pmod{18}$$

Thus, the remainder is 13.